

Lab 25: Radial Basis Function (RBF) Network

Aim

To implement an **RBF Network** with **K-Means selected centres** for classification using the Iris dataset.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Use **K-Means clustering** to select RBF centres.
5. Calculate the Gaussian RBF activation for each data point.
6. Transform the original features into RBF features.
7. Train a linear classifier using the transformed features.
8. Predict the test data and calculate the accuracy.

Program

```
# Lab 25: Radial Basis Function (RBF) Network
```

```
import numpy as np

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.cluster import KMeans
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
```

```
# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Select RBF centres using K-Means
```

```
kmeans = KMeans(
    n_clusters=10,
    random_state=42,
    n_init=10
)
```

```
kmeans.fit(X_train)
```

```
centers = kmeans.cluster_centers_
```

```
# Gaussian RBF transformation
```

```
def rbf_transform(X, centers, gamma=0.5):
    return np.exp(
        -gamma * np.sum(
            (X[:, np.newaxis] - centers) ** 2,
            axis=2
        )
    )
```

```
# Transform training and testing data
```

```
X_train_rbf = rbf_transform(X_train, centers)
```

```
X_test_rbf = rbf_transform(X_test, centers)
```

```
# Train classifier
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train_rbf, y_train)
```

```
# Make predictions
```

```
y_pred = model.predict(X_test_rbf)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Sample Output

Accuracy: 1.0

The accuracy may vary slightly depending on the train-test split and RBF parameters.

Result

Thus, the **Radial Basis Function (RBF) Network was successfully implemented** using K-Means selected centres, and the Iris dataset was classified using Gaussian RBF-transformed features.